WEEK 6 ASSIGNMENT — FORMAL EVIDENCE REPORT

Dockerization & Verification Evidence Report

Information Systems Agile Project Management with DevOps (Class: IS 2026) | Lecturer: Eka Parima Saragih, S.Kom., M.Kom.

Course: Agile Project Management with DevOps **Deliverable:** Week 6 Docker Evidence Report

Project Title: Mocha Topia (Coffee Shop Website) **Submission Filename:** `W6_Group04_MochaTopia.pdf`

Section A: Group Identity

Field	Group Response
Group Number	Group 04
Project Title	Mocha Topia — Coffee Shop Web Application
Group Members & IDs	1. **Faiz** (Student ID: `012202500030`) 2. **Immanuel** (Student ID: `012202500017`) 3. **Dwisasono** (Student ID: `012202500064`)
Repository URL	`https://github.com/masfaiz1307-pixel/mocha-topia`
Git Working Branch	`chore/dockerize-project` (merged into `master`)
Demonstration Video URL	`https://drive.google.com/file/d/15GMW1KzOlkNakVKkog_W5PYNIK5CJhup/view?usp=drivesdk` (Max 20 min)

Section B: Project Baseline

Before introducing Docker, the project baseline was tested to confirm that all core functionality operates cleanly without container errors.

Baseline Item	Group Response
Project name and repository	Mocha Topia (`https://github.com/masfaiz1307-pixel/mocha-topia`)
Main user or purpose	Coffee shop customers browsing drink/food menus, placing orders, and contacting the café
Technology stack	HTML5, CSS3, JavaScript (ES6), jQuery 3.6.0 (CDN)
Normal start command	Open `index.html` directly in a modern web browser
Application port	N/A locally (static files); served on container port 80 via Nginx
Required environment variables	None (client-side web application)
Supporting services	None (browser `localStorage` persists accounts and cart state)

Baseline Item	Group Response
Core function for verification	1. User Registration, 2. Login Authentication, 3. Menu Browsing & Cart Management

Section C: Container Design

The application architecture was evaluated to create the most minimal, secure, and performant container image:

Design Element	Decision	Technical Rationale
Base Image	`nginx:1.27-alpine`	Alpine Linux footprint (~5MB) yields a production image of only **77.9 MB**. Node.js or Ubuntu would introduce unnecessary 150MB+ runtimes for static assets.
Image Name & Tag	`mocha-topia:week6`	Complies with the required `<group-name>/<project-name>:week6` tag convention.
Container Port	Port `80` → Host `8080`	Container listens on default HTTP port 80; mapped to host port 8080 to prevent conflicts with host services.
Environment Variables	`.env.example` template	Zero-configuration design; no sensitive secrets or keys are hardcoded or baked into the image.
Exclusions	`.dockerignore`	Excludes `.git`, `.env`, IDE directories, and documentation from build context.

Dockerfile

```
FROM nginx:1.27-alpine
WORKDIR /usr/share/nginx/html
RUN rm -rf ./*
COPY . .
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

Section D: Build Evidence

Build Command: `docker build -t mocha-topia:week6 .` (or `docker compose up --build -d`)

Image Listing Verification (`docker image ls`):

```
PS C:\Users\Lenovo\OneDrive\Documents\mocha-topia> docker image ls mocha-topia-app
REPOSITORY        TAG       IMAGE ID       CREATED        SIZE
mocha-topia-app   latest    11e3760d75f0   18 hours ago   77.9MB

PS C:\Users\Lenovo\OneDrive\Documents\mocha-topia> docker image ls mocha-topia
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
mocha-topia   week6     439512f21080   1 hour ago     77.9MB
```

Build Explanation: All 7 layers completed without warnings. The resulting image size is 77.9 MB, creating a fast-pulling, lightweight container.

Section E: Runtime Evidence

Run Command: `docker compose up -d`

Service Status (`docker compose ps`):

```
NAME              IMAGE             COMMAND                  SERVICE   STATUS         PORTS
mocha-topia-app   mocha-topia-app   "/docker-entrypoint.…"   app       Up (healthy)   0.0.0.0:8080->80/tcp
```

Container Logs (`docker compose logs`):

```
mocha-topia-app  | /docker-entrypoint.sh: Configuration complete; ready for start up
mocha-topia-app  | 2026/10/18 04:10:26 [notice] 1#1: using the "epoll" event method
mocha-topia-app  | 2026/10/18 04:10:26 [notice] 1#1: nginx/1.27.5
mocha-topia-app  | 2026/10/18 04:10:26 [notice] 1#1: start worker processes
```

HTTP Server Response (`curl.exe -I http://localhost:8080`):

```
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Sun, 18 Oct 2026 04:54:19 GMT
Content-Type: text/html
Content-Length: 1133
Connection: keep-alive
```

Section F: Functional Verification

Check ID	Verification Action	Expected Result	Actual Result / Evidence	Status
V-01	Container & Service Status	Status Up on port 8080	Container active; 0.0.0.0:8080->80/tcp	✅ PASS
V-02	HTTP Availability	Status Code 200 OK returned	HTTP/1.1 200 OK from Nginx 1.27.5	✅ PASS
V-03	Core Feature: User Registration	User profile saved in localStorage	Validation passes, account registered	✅ PASS
V-04	Core Feature: User Login	Authentication succeeds	Redirected to main dashboard session	✅ PASS
V-05	Core Feature: Menu & Cart	Items added and totals calculated	Cart dynamically updates quantities and prices	✅ PASS
V-06	Container Restart Lifecycle	Stack restarts cleanly	`docker compose restart` preserves state	✅ PASS

Visual Verification Proof (Screenshots of Containerized Execution):

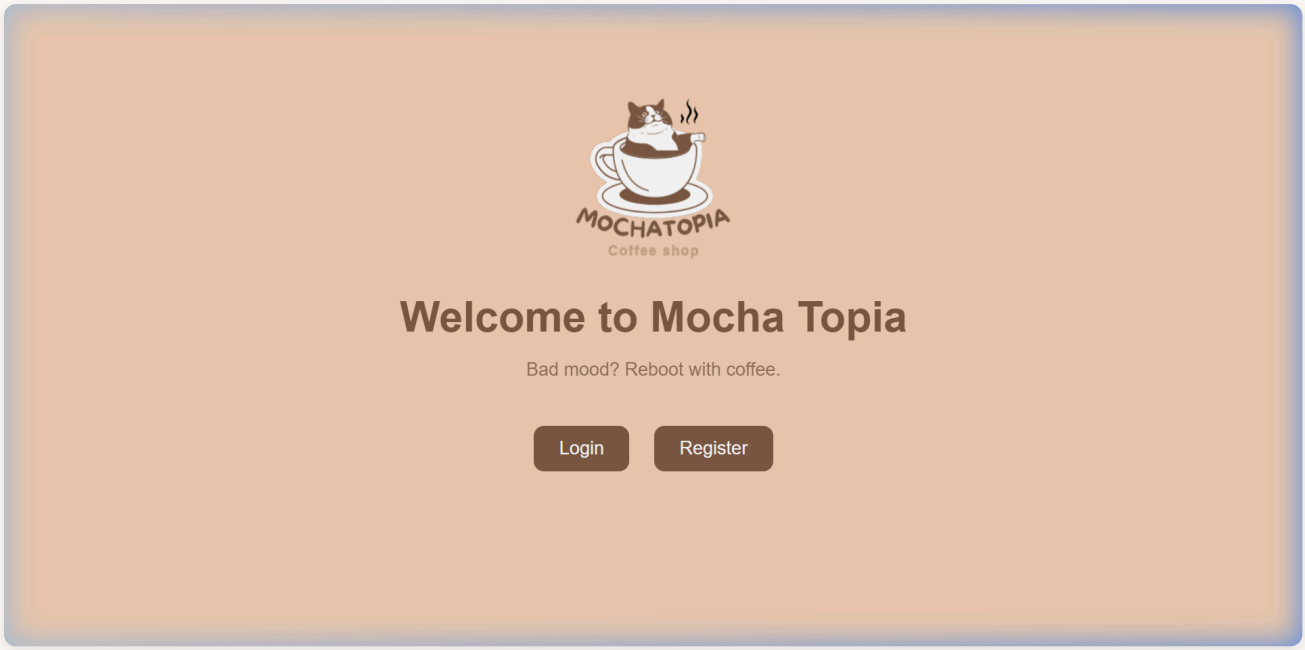


Figure 1: Landing Page served by Nginx container at http://localhost:8080

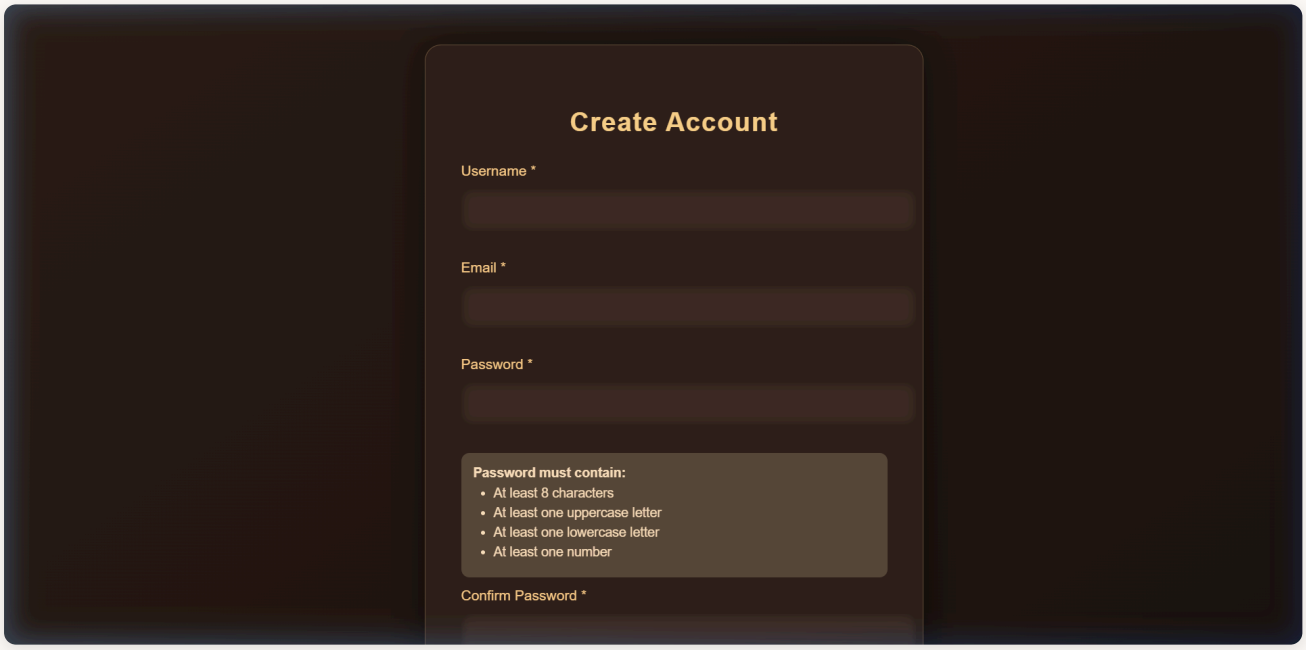


Figure 2: User registration page with client-side form validation

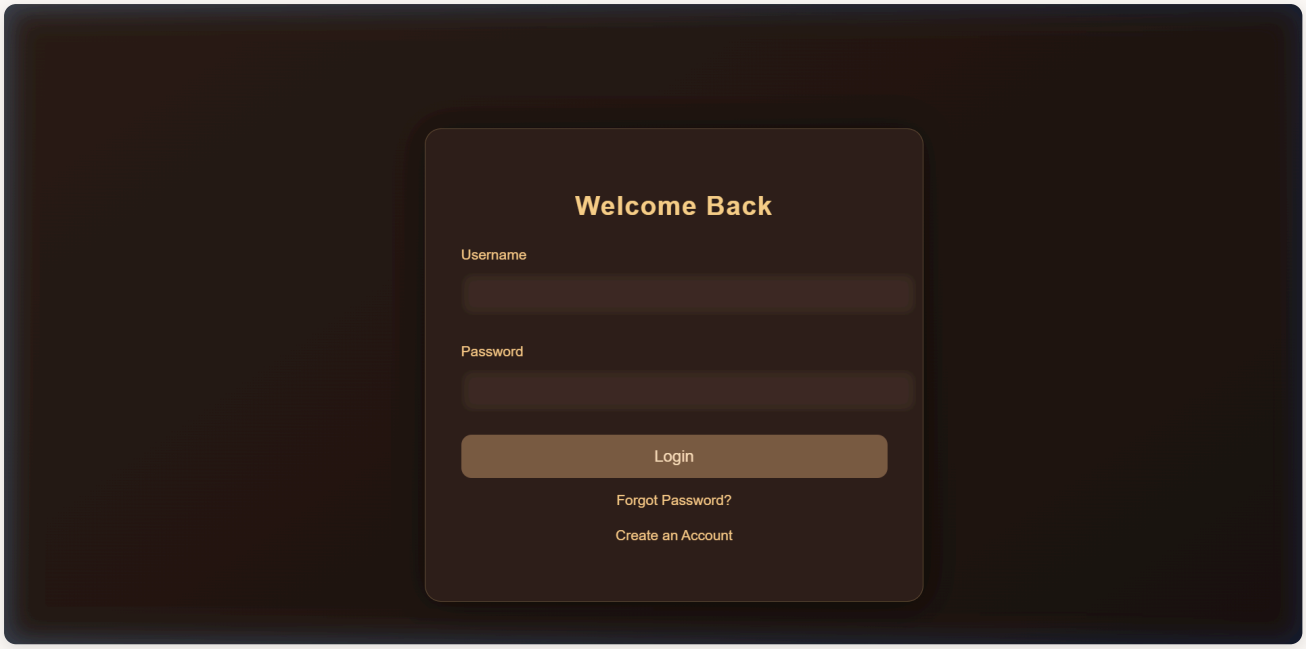


Figure 3: Login authentication verifying against localStorage credentials

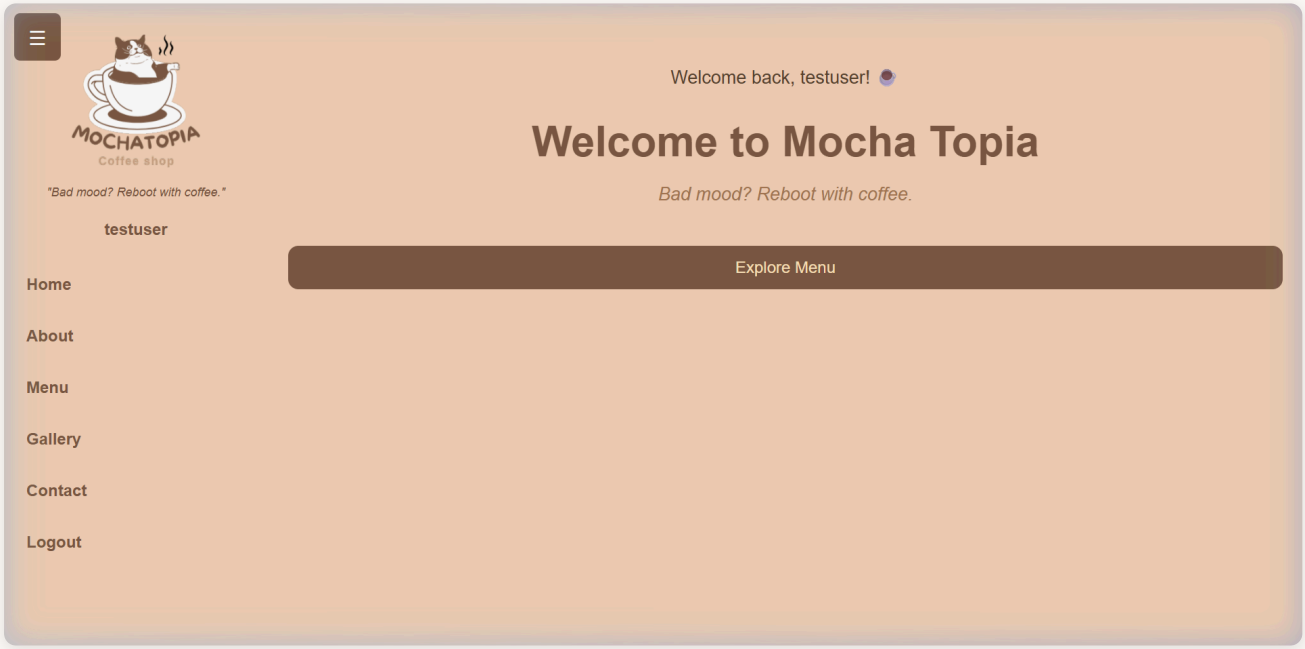


Figure 4: Main coffee shop dashboard, frameset navigation, and menu catalog

Section G: Handover Result

A member from another group conducted the peer handover test to prove that the instructions in `README.md` are reproducible from a fresh clone.

Handover Record	Response
Verifier Name	`Fayyaz`
Verifier Group Number	`Group 7`

Handover Record	Response
Device / Operating System	Windows 11 / Docker Desktop v28.x
Fresh Clone Completed?	**Yes** (Cloned branch `chore/dockerize-project`)
Image Build Successful?	**Yes** (Built cleanly in ~15 seconds)
Application Start Successful?	**Yes** (Accessible at `http://localhost:8080`)
Core Functions Verified?	**Yes** (Registration, Login, and Cart tested)
Problems Found & Corrections Made	PowerShell `curl` alias caused drive lookup issue. Documented using `curl.exe` or direct browser access in README.

Section H: Troubleshooting Reflection

Diagnosis Step	Details
Real Problem	`Bind for 0.0.0.0:8080 failed: port is already allocated` during container startup.
Root Cause	A previous container was started with standalone `docker run`. Running `docker compose down` later only cleaned up Compose-managed containers, leaving the standalone container running and holding port 8080.
Diagnostic Evidence	`docker ps -a` revealed container ID `f73254a70c8a` active on port 8080 under name `mocha-topia-app`.
Solution Applied	Ran `docker stop mocha-topia-app && docker rm mocha-topia-app`, then restarted cleanly with `docker compose up -d`.
Lesson Learned	Do not mix standalone `docker run` and `docker compose` workflows. Always standardize on Docker Compose across all developer instructions.

Section I: Individual Contribution Table

Contribution breakdown for all team members as required by Table 5.1:

Member / Student ID	Specific Contribution	Commit / PR Evidence	Concept the Member Can Explain
Faiz `012202500030`	Authored `Dockerfile`, customized `nginx.conf` Gzip & cache rules, solved port allocation conflict, and verified browser workflows.	Commits: `91dd342`, `1fcd381` PR: `chore/dockerize-project`	Base image selection (Nginx Alpine vs Node.js), build caching, and foreground process execution (`daemon off;`).
Immanuel `012202500017`	Configured `compose.yaml` service definitions, host-container port mapping, and constructed `.dockerignore` rules.	Commits: `91dd342`, `f37f0d7` PR: `chore/dockerize-project`	Port binding (`8080:80`), Compose service definitions, and security benefits of excluding secrets via `.dockerignore`.

Member / Student ID	Specific Contribution	Commit / PR Evidence	Concept the Member Can Explain
Dwisasono `012202500064`	Led handover testing with peer group, executed test matrix (V-01 through V-06), and documented troubleshooting procedures.	Commits: `f37f0d7`, `1fcd381` PR: `chore/dockerize-project`	Handover reproducibility criteria, container inspection commands (`ps`, `logs`, `inspect`), and lifecycle teardown.